

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		18.8 g (3) if answer incorrect credit each correct step in one of two possible methods (ecf possible throughout) method 1 $n(K) = \frac{15.6}{39} = 0.4 \text{ mol} \quad (1)$ $n(K_2O) = \frac{0.4}{2} = 0.2 \text{ mol} \quad (1)$ $\text{mass } K_2O = 0.2 \times 94 = 18.8 \text{ g} \quad (1)$ method 2 $M_r(K_2O) = 94 / \text{mass of } 156 \text{ (for K)}(1)$ (156 g K produces) 188 g $K_2O \quad (1)$ 15.6 g K produces 18.8 g $K_2O \quad (1)$		3		3	3	
		(ii)		$3.0 \times 10^{22} \text{ (2)}$ accept 3×10^{22} if answer incorrect award 1 mark for 0.30×10^{23}		2		2	2	
				Question 9 total	6	5	0	11	5	3